Chapter 17

Lie Group Homomorphisms and Their Differentials

Road map

A homomorphism of matrix Lie groups carries one-parameter subgroups to one-parameter subgroups. This simple observation produces a unique real linear map between the corresponding Lie algebras. We prove carefully that the induced map commutes with exponentials, respects conjugation and Lie brackets, and is computed by differentiating the group homomorphism along e^{tX}. We then revisit the adjoint representation, derive Hadamard's lemma and the Heisenberg evolution series, show that a homomorphism on a connected group is determined by its Lie-algebra map, and prove that every continuous matrix-group homomorphism is automatically smooth. The chapter ends by relating the centre of a connected Lie group to the centre of its Lie algebra.

17.1 Group and Lie-algebra homomorphisms

Definition 17.1: Lie group homomorphism

Let G and H be linear Lie groups. A map

$$\Phi : G \longrightarrow H$$

is a *Lie group homomorphism* if

1. it is a group homomorphism:

$$\Phi(AB) = \Phi(A)\Phi(B) \qquad (A, B \in G);$$

2. it is continuous.

Continuity is enough in the definition: we shall prove later in this chapter that such a homomorphism is automatically smooth.

Definition 17.2: Lie-algebra homomorphism

Let $\mathfrak{g}$ and $\mathfrak{h}$ be Lie algebras. A map

$$\psi : \mathfrak{g} \longrightarrow \mathfrak{h}$$

is a *Lie-algebra homomorphism* if it is linear and

$$\psi([X, Y]) = [\psi(X), \psi(Y)] \qquad (X, Y \in \mathfrak{g}). \tag{17.1}$$

The goal is to pass canonically from Φ to ψ.

17.2 Differentiating a Lie group homomorphism

Theorem 17.1: The homomorphism–exponential correspondence

Let G, H be linear Lie groups, with

$$\mathfrak{g} = \mathrm{Lie}(G), \qquad \mathfrak{h} = \mathrm{Lie}(H),$$

and let $\Phi : G \to H$ be a Lie group homomorphism. There is a unique real-linear map

$$\psi : \mathfrak{g} \longrightarrow \mathfrak{h}$$

such that

$$\boxed{\Phi(\mathrm{e}^X) = \mathrm{e}^{\psi(X)} \qquad (X \in \mathfrak{g}).} \tag{17.2}$$

Equivalently, the following diagram commutes:

$$\begin{tikzcd}
\mathfrak{g} \arrow[r, "\psi"] \arrow[d, "\exp"'] & \mathfrak{h} \arrow[d, "\exp"] \\
G \arrow[r, "\Phi"'] & H
\end{tikzcd}$$

The map is computed by

$$\boxed{\psi(X) = \frac{\mathrm{d}}{\mathrm{d}t}\Phi(\mathrm{e}^{tX})\Big|_{t=0}.} \tag{17.3}$$

Moreover,

$$\psi(AXA^{-1}) = \Phi(A)\psi(X)\Phi(A)^{-1}, \qquad A \in G,\ X \in \mathfrak{g}, \tag{17.4}$$

$$\psi([X, Y]) = [\psi(X), \psi(Y)], \qquad X, Y \in \mathfrak{g}. \tag{17.5}$$

Thus ψ is in fact a Lie-algebra homomorphism.

证明. **Step 1: construct $\psi(X)$ from a one-parameter subgroup.** Fix $X \in \mathfrak{g}$. The map

$$\mathbb{R} \longrightarrow H, \qquad t \longmapsto \Phi(\mathrm{e}^{tX})$$

is continuous. It is a homomorphism because

$$\begin{aligned}\Phi(e^{(t+s)X}) &= \Phi(e^{tX}e^{sX})\\ &= \Phi(e^{tX})\Phi(e^{sX}).\end{aligned}$$

It is therefore a one-parameter subgroup of H. By the classification of one-parameter matrix subgroups, there is a unique $Z \in \mathfrak{h}$ such that

$$\Phi(e^{tX}) = e^{tZ} \qquad (t \in \mathbb{R}).$$

Define

$$\psi(X) := Z.$$

Setting $t = 1$ gives (17.2). Differentiating at $t = 0$ gives (17.3).

Step 2: homogeneity over $\mathbb{R}$. For $s, t \in \mathbb{R}$,

$$e^{t\psi(sX)} = \Phi(e^{tsX}) = e^{ts\psi(X)}.$$

Differentiating with respect to t at 0 gives

$$\psi(sX) = s\psi(X).$$

This proves real homogeneity. Even if the groups consist of complex matrices, the natural Lie algebras here are real unless extra complex linearity is separately established.

Step 3: additivity. Use the Lie–Trotter product formula:

$$e^{t(X+Y)} = \lim_{m\to\infty} \left(e^{tX/m}e^{tY/m}\right)^m.$$

Continuity of Φ and its homomorphism property give

$$\begin{aligned}e^{t\psi(X+Y)} &= \Phi(e^{t(X+Y)})\\ &= \lim_{m\to\infty} \left(\Phi(e^{tX/m})\Phi(e^{tY/m})\right)^m\\ &= \lim_{m\to\infty} \left(e^{t\psi(X)/m}e^{t\psi(Y)/m}\right)^m\\ &= e^{t(\psi(X)+\psi(Y))}.\end{aligned}$$

Differentiating at $t = 0$ yields

$$\psi(X + Y) = \psi(X) + \psi(Y).$$

Together with Step 2, this proves that ψ is real linear.

Step 4: uniqueness. Suppose another real-linear map $\psi' : \mathfrak{g} \to \mathfrak{h}$ satisfies $\Phi(e^X) = e^{\psi'(X)}$. Replacing X by tX,

$$e^{t\psi'(X)} = \Phi(e^{tX}) = e^{t\psi(X)}.$$

Differentiating at $t = 0$ gives $\psi'(X) = \psi(X)$ for every X.

Step 5: compatibility with conjugation. For $A \in G$, $X \in \mathfrak{g}$, and $t \in \mathbb{R}$,

$$\begin{aligned} e^{t\psi(AXA^{-1})} &= \Phi(e^{tAXA^{-1}}) \\ &= \Phi(Ae^{tX}A^{-1}) \\ &= \Phi(A)\Phi(e^{tX})\Phi(A)^{-1} \\ &= \Phi(A)e^{t\psi(X)}\Phi(A)^{-1}. \end{aligned}$$

Differentiation at $t = 0$ gives (17.4).

Step 6: preservation of brackets. For $X, Y \in \mathfrak{g}$,

$$[X, Y] = \frac{\mathrm{d}}{\mathrm{d}t}\left(e^{tX}Ye^{-tX}\right)\bigg|_{t=0}.$$

Because ψ is linear, it commutes with differentiation of this finite-dimensional curve:

$$\begin{aligned} \psi([X, Y]) &= \frac{\mathrm{d}}{\mathrm{d}t}\psi(e^{tX}Ye^{-tX})\bigg|_{t=0} \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\left(\Phi(e^{tX})\psi(Y)\Phi(e^{tX})^{-1}\right)\bigg|_{t=0} \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\left(e^{t\psi(X)}\psi(Y)e^{-t\psi(X)}\right)\bigg|_{t=0} \\ &= [\psi(X), \psi(Y)]. \end{aligned}$$

This is (17.5). ■

What the theorem does and does not say

Every continuous Lie group homomorphism differentiates to a Lie-algebra homomorphism. The converse integration problem is subtler: a Lie-algebra homomorphism need not descend to a prescribed non-simply-connected group. Existence becomes clean when the source group is simply connected; the covering-group chapters will explain the missing global information.

Example 17.1: The determinant differentiates to the trace

Take

$$\Phi = \det : \mathrm{GL}(n, \mathbb{C}) \longrightarrow \mathbb{C}^\times.$$

The target has Abelian Lie algebra $\mathbb{C}$, written additively. Since

$$\det(e^{tX}) = e^{t\,\mathrm{tr}\,X},$$

the induced Lie-algebra homomorphism is

$$\psi(X) = \mathrm{tr}\,X.$$

The group-level kernel is $\mathrm{SL}(n, \mathbb{C})$, and the infinitesimal kernel is $\mathfrak{sl}(n, \mathbb{C})$.

17.3 The adjoint homomorphism revisited

Let $G \leq \mathrm{GL}(n, \mathbb{K})$ be a linear Lie group with $\mathfrak{g} = \mathrm{Lie}(G)$. For $A \in G$, define

$$\mathrm{Ad}_A : \mathfrak{g} \longrightarrow \mathfrak{g}, \qquad \mathrm{Ad}_A(X) = AXA^{-1}.$$

This is well defined because conjugation by A preserves $\mathfrak{g}$.

Proposition 17.1: Each Ad_A is a Lie-algebra automorphism

For $A \in G$, the map Ad_A is linear, bijective, and preserves brackets:

$$\mathrm{Ad}_A([X, Y]) = [\mathrm{Ad}_A X, \mathrm{Ad}_A Y].$$

证明. Linearity follows from

$$\begin{aligned} \mathrm{Ad}_A(\lambda_1 X_1 + \lambda_2 X_2) &= A(\lambda_1 X_1 + \lambda_2 X_2)A^{-1} \\ &= \lambda_1 A X_1 A^{-1} + \lambda_2 A X_2 A^{-1}. \end{aligned}$$

Its inverse is $\mathrm{Ad}_{A^{-1}}$, so it is bijective. Finally,

$$\begin{aligned} \mathrm{Ad}_A([X, Y]) &= A(XY - YX)A^{-1} \\ &= (AXA^{-1})(AYA^{-1}) - (AYA^{-1})(AXA^{-1}) \\ &= [\mathrm{Ad}_A X, \mathrm{Ad}_A Y]. \end{aligned}$$

■

Let $\mathrm{GL}(\mathfrak{g})$ denote the group of invertible real-linear maps $\mathfrak{g} \to \mathfrak{g}$. If $d = \dim_{\mathbb{R}} \mathfrak{g} = \dim G$, then after choosing a basis

$$\mathrm{GL}(\mathfrak{g}) \cong \mathrm{GL}(d, \mathbb{R}),$$

so it is itself a matrix Lie group.

Theorem 17.2: The group adjoint representation

The map

$$\mathrm{Ad} : G \longrightarrow \mathrm{GL}(\mathfrak{g}), \qquad A \longmapsto \mathrm{Ad}_A \tag{17.6}$$

is a Lie group homomorphism.

证明. For $A, B \in G$ and $X \in \mathfrak{g}$,

$$\begin{aligned} \mathrm{Ad}_{AB}(X) &= (AB)X(AB)^{-1} \\ &= A(BXB^{-1})A^{-1} \\ &= \mathrm{Ad}_A(\mathrm{Ad}_B X). \end{aligned}$$

Thus

$$\mathrm{Ad}_{AB} = \mathrm{Ad}_A \mathrm{Ad}_B .$$

For continuity, if $A_m \to A$ in G, then

$$A_m X A_m^{-1} \longrightarrow A X A^{-1}$$

for every X, because matrix multiplication and inversion are continuous. In finite dimensions this is exactly continuity of $\mathrm{Ad}_{A_m} \to \mathrm{Ad}_A$ as linear operators. ■

The Lie algebra of $\mathrm{GL}(\mathfrak{g})$ is $\mathfrak{gl}(\mathfrak{g})$. Applying the preceding theorem on differentiating homomorphisms gives an associated Lie-algebra homomorphism

$$\mathrm{ad} : \mathfrak{g} \longrightarrow \mathfrak{gl}(\mathfrak{g}).$$

For $X, Y \in \mathfrak{g}$,

$$\begin{aligned}
\mathrm{ad}_X(Y) &= \frac{\mathrm{d}}{\mathrm{d}t} \mathrm{Ad}_{\mathrm{e}^{tX}}(Y)\Big|_{t=0} \\
&= \frac{\mathrm{d}}{\mathrm{d}t} \left(\mathrm{e}^{tX} Y \mathrm{e}^{-tX}\right)\Big|_{t=0} \\
&= XY - YX \\
&= [X, Y].
\end{aligned}$$

This recovers exactly the infinitesimal adjoint action defined in the preceding chapter.

$$\begin{tikzcd}
G \arrow[rr, "\mathrm{Ad}"] & & \mathrm{GL}(\mathfrak{g}) \\
\mathfrak{g} \arrow[u, "\exp"] \arrow[rr, "\mathrm{ad}"] & & \mathfrak{gl}(\mathfrak{g}) \arrow[u, "\exp"']
\end{tikzcd}$$

图 **17.1:** The adjoint representation commutes with exponential: $\mathrm{Ad}_{\mathrm{e}^X} = \mathrm{e}^{\mathrm{ad}_X}$.

17.4 Hadamard's lemma

Applying the exponential compatibility $\Phi(\mathrm{e}^X) = \mathrm{e}^{\psi(X)}$ to $\Phi = \mathrm{Ad}$ and $\psi = \mathrm{ad}$ gives

$$\mathrm{Ad}_{\mathrm{e}^X} = \mathrm{e}^{\mathrm{ad}_X}. \tag{17.7}$$

Acting on $Y \in \mathfrak{g}$, we obtain the following formula.

Theorem 17.3: Hadamard's lemma

For matrices X, Y, or more generally whenever the relevant exponential and adjoint action are defined,

$$\boxed{\mathrm{e}^X Y \mathrm{e}^{-X} = Y + [X, Y] + \frac{1}{2!}[X, [X, Y]] + \frac{1}{3!}[X, [X, [X, Y]]] + \cdots.} \tag{17.8}$$

Equivalently,

$$e^{X} Y e^{-X} = \sum_{k=0}^{\infty} \frac{1}{k!} \operatorname{ad}_X^k(Y).$$

证明. Equation (17.7) gives

$$e^{X} Y e^{-X} = \operatorname{Ad}_{e^X}(Y) = e^{\operatorname{ad}_X}(Y).$$

Expand the exponential of the linear operator ad_X:

$$e^{\operatorname{ad}_X}(Y) = \sum_{k=0}^{\infty} \frac{1}{k!} \operatorname{ad}_X^k(Y).$$

The first iterates are

$$\operatorname{ad}_X^0(Y) = Y, \qquad \operatorname{ad}_X^1(Y) = [X, Y], \qquad \operatorname{ad}_X^2(Y) = [X, [X, Y]],$$

and so on, which is (17.8). ■

Example 17.2: Heisenberg evolution of an observable

Let $\mathcal{H}$ be a Hilbert space, H a Hamiltonian, and A an observable. In finite dimensions all operators are bounded, so every series below converges without domain subtleties. The Heisenberg-picture time dependence is

$$A(t) = e^{iHt} A e^{-iHt} = \operatorname{Ad}_{e^{iHt}} A.$$

Hadamard's lemma gives

$$\begin{aligned} A(t) &= e^{\operatorname{ad}_{iHt}} A \\ &= \sum_{k=0}^{\infty} \frac{(it)^k}{k!} \underbrace{[H, [H, \dots, [H, A] \dots]]}_{k \text{ brackets}}. \end{aligned} \tag{17.9}$$

Differentiating directly,

$$\frac{dA(t)}{dt} = i[H, A(t)],$$

the Heisenberg equation of motion in units $\hbar = 1$.
If A is an eigenoperator of ad_H,

$$[H, A] = \omega A, \qquad \omega \in \mathbb{C},$$

then induction gives

$$\operatorname{ad}_H^k(A) = \omega^k A.$$

Therefore

$$A(t) = \sum_{k=0}^{\infty} \frac{(it\omega)^k}{k!} A = e^{i\omega t} A.$$

17.5 Connected groups are determined infinitesimally

Theorem 17.4: Uniqueness on a connected source

Let G, H be linear Lie groups and assume that G is connected. Let

$$\Phi_1, \Phi_2 : G \longrightarrow H$$

be Lie group homomorphisms, with induced Lie-algebra homomorphisms

$$\psi_1, \psi_2 : \mathfrak{g} \longrightarrow \mathfrak{h}.$$

If $\psi_1 = \psi_2$, then

$$\Phi_1 = \Phi_2.$$

证明. Every $A \in G$ is a finite product of exponentials:

$$A = \mathrm{e}^{X_1} \cdots \mathrm{e}^{X_m}, \qquad X_j \in \mathfrak{g}.$$

Using the homomorphism property and exponential compatibility,

$$\begin{aligned}
\Phi_1(A) &= \Phi_1(\mathrm{e}^{X_1}) \cdots \Phi_1(\mathrm{e}^{X_m}) \\
&= \mathrm{e}^{\psi_1(X_1)} \cdots \mathrm{e}^{\psi_1(X_m)} \\
&= \mathrm{e}^{\psi_2(X_1)} \cdots \mathrm{e}^{\psi_2(X_m)} \\
&= \Phi_2(\mathrm{e}^{X_1}) \cdots \Phi_2(\mathrm{e}^{X_m}) \\
&= \Phi_2(A).
\end{aligned}$$

■

Why connectedness appears

The differential sees only the identity component. On a disconnected group, two homomorphisms can agree on G°, hence have the same Lie-algebra map, but differ on other components. Connectedness eliminates this discrete ambiguity.

17.6 Continuity implies smoothness

Theorem 17.5: Automatic smoothness

Every continuous homomorphism between linear Lie groups is smooth (indeed, real analytic).

证明. Let $\Phi : G \to H$ be continuous, and let $\psi : \mathfrak{g} \to \mathfrak{h}$ be its induced real-linear map. Fix $A \in G$. If B is sufficiently close to A, then

$$A^{-1}B$$

is close to I and lies in the local exponential neighbourhood of G. Hence

$$A^{-1}B = \mathrm{e}^X, \qquad X = \log(A^{-1}B) \in \mathfrak{g}.$$

Therefore

$$\begin{aligned}\Phi(B) &= \Phi(Ae^X) \\ &= \Phi(A)\Phi(e^X) \\ &= \Phi(A)e^{\psi(X)} \\ &= \Phi(A)\exp\bigl(\psi(\log(A^{-1}B))\bigr). \end{aligned} \tag{17.10}$$

Near A, this is a composition of smooth maps:

$$B \longmapsto A^{-1}B \longmapsto \log(A^{-1}B) \longmapsto \psi(\log(A^{-1}B)) \longmapsto \exp(\psi(\log(A^{-1}B))) \longmapsto \Phi(A)\exp(\cdots).$$

Thus Φ is smooth in a neighbourhood of every A, hence smooth on all of G. Since the matrix exponential and local logarithm are analytic power series, the same argument gives real analyticity. ∎

17.7 Centres of connected groups and Lie algebras

Theorem 17.6: Centre correspondence

If G is a connected linear Lie group and $\mathfrak{g} = \mathrm{Lie}(G)$, then

$$\boxed{Z(\mathfrak{g}) = \mathrm{Lie}(Z(G)).} \tag{17.11}$$

证明. We prove a chain of equivalent conditions for $X \in \mathfrak{g}$.

First,

$$\begin{aligned} X \in Z(\mathfrak{g}) &\iff [Y, X] = 0 \quad \text{for every } Y \in \mathfrak{g} \\ &\iff \mathrm{ad}_Y(X) = 0 \quad \text{for every } Y \in \mathfrak{g}. \end{aligned}$$

This is equivalent to

$$e^{t\,\mathrm{ad}_Y} X = X \qquad \text{for every } Y \in \mathfrak{g},\ t \in \mathbb{R}.$$

The forward implication is immediate from the exponential series. For the reverse implication, differentiate at $t = 0$. It is important that the equality holds for *all* t; the isolated equation $e^T = I$ would not by itself imply $T = 0$.

Using $e^{t\,\mathrm{ad}_Y} = \mathrm{Ad}_{e^{tY}}$, we have

$$X \in Z(\mathfrak{g}) \iff \mathrm{Ad}_{e^{tY}} X = X \quad (Y \in \mathfrak{g},\ t \in \mathbb{R}).$$

Because G is connected, every $g \in G$ is a finite product

$$g = e^{Y_1} \cdots e^{Y_m}.$$

The adjoint map is a group homomorphism, so invariance under each exponential factor is equivalent to invariance under every $g \in G$:

$$X \in Z(\mathfrak{g}) \iff \mathrm{Ad}_g X = X \quad (g \in G).$$

In matrix notation this is

$$gXg^{-1} = X \quad \iff \quad gX = Xg \qquad (g \in G).$$

If $gX = Xg$, then g commutes with every power X^k, and hence with the exponential series:

$$g\mathrm{e}^{tX} = \mathrm{e}^{tX}g \qquad (g \in G,\ t \in \mathbb{R}).$$

Conversely, if this exponential commutation holds for every t, differentiation at $t = 0$ gives $gX = Xg$. Therefore

$$\begin{aligned} X \in Z(\mathfrak{g}) &\iff \mathrm{e}^{tX} \in Z(G) \quad \text{for every } t \in \mathbb{R} \\ &\iff X \in \mathrm{Lie}(Z(G)). \end{aligned}$$

This proves (17.11). ■

Connectedness is essential

Without connectedness, commuting with every infinitesimal direction only guarantees commutation with G°. A disconnected component can act nontrivially by conjugation while remaining invisible to $\mathrm{Lie}(G)$.

What to retain from this chapter

1. A Lie group homomorphism is a continuous group homomorphism. It is automatically smooth.
2. Every $\Phi : G \to H$ induces a unique real-linear map
$$\psi : \mathrm{Lie}(G) \to \mathrm{Lie}(H)$$
satisfying
$$\Phi(\mathrm{e}^X) = \mathrm{e}^{\psi(X)}, \qquad \psi(X) = \frac{\mathrm{d}}{\mathrm{d}t}\Phi(\mathrm{e}^{tX})\Big|_{t=0}.$$
3. The induced map respects conjugation and brackets:
$$\psi(AXA^{-1}) = \Phi(A)\psi(X)\Phi(A)^{-1}, \qquad \psi([X,Y]) = [\psi(X), \psi(Y)].$$
Hence ψ is a Lie-algebra homomorphism.
4. The group adjoint representation
$$\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$$
differentiates to
$$\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g}), \qquad \mathrm{ad}_X(Y) = [X,Y].$$
5. $$\mathrm{Ad}_{\mathrm{e}^X} = \mathrm{e}^{\mathrm{ad}_X},$$
which yields Hadamard's lemma
$$\mathrm{e}^X Y \mathrm{e}^{-X} = Y + [X,Y] + \frac{1}{2!}[X,[X,Y]] + \cdots.$$
6. If the source group is connected, two Lie group homomorphisms with the same induced Lie-algebra homomorphism are equal.
7. For a connected linear Lie group,
$$Z(\mathrm{Lie}(G)) = \mathrm{Lie}(Z(G)).$$